

Brain Tumour MRI Image Classification

A deep learning-based approach to detect and classify brain tumours from MRI scans using AI-powered medical imaging analysis.

Developed By: Yogapriya

Domain: Medical Imaging
Deep Learning & AI

Technologies: Python,

TensorFlow/Keras, Streamlit

Dataset: Brain MRI Images (4
Classes)



Problem Statement & Objective

The Challenge

Brain tumour diagnosis through MRI images is time-consuming and prone to human error. Manual interpretation by radiologists can delay critical treatment decisions.

Our Solution

Develop an AI-powered system that classifies MRI brain images into tumour types, supporting early diagnosis and faster decision-making to assist radiologists.

01

Build Custom CNN

Develop convolutional neural network architecture

02

Apply Transfer Learning

Leverage pre-trained models for classification

03

Compare Performance

Evaluate and select best-performing model

04

Deploy via Streamlit

Create accessible web application interface



Dataset Overview

Four Tumour Categories

- ☒ Glioma
- ☐ Meningioma
- ☐ Pituitary
- ☐ No Tumour

Dataset Characteristics

Source: Brain MRITumourDataset

Image Size: All images resized to 224×224 pixels

Balance: Equal distribution across all four classes

Normalisation

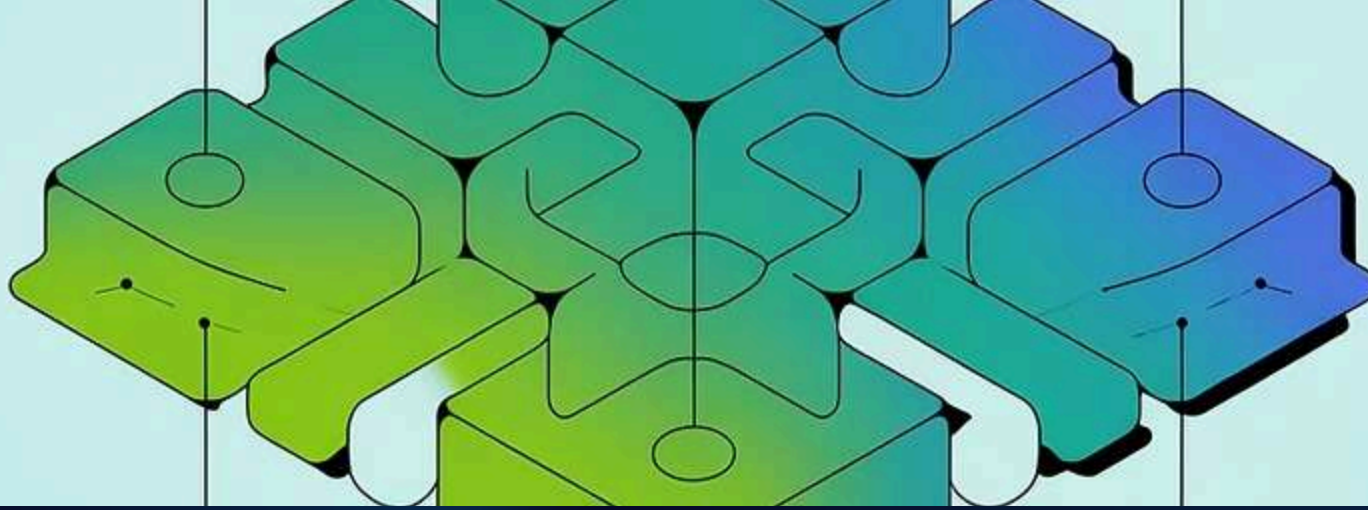
Pixel values scaled to 0-1 range
for consistent model input

Image Resizing

Standardised dimensions
ensure uniform processing
across dataset

Data Augmentation

Rotation, flip, zoom, and
brightness adjustments
enhance model robustness



Model Building & Training



Custom CNN Model

- Convolution, Pooling, Dropout, Dense layers
- Batch Normalisation and ReLU activation
- Optimizer: Adam | Loss: Categorical Crossentropy



Transfer Learning Models

- Pre-trained: ResNet50, InceptionV3, MobileNetV2
- Customised layers for 4-class classification
- Fine-tuned top layers for enhanced accuracy

Training Strategy

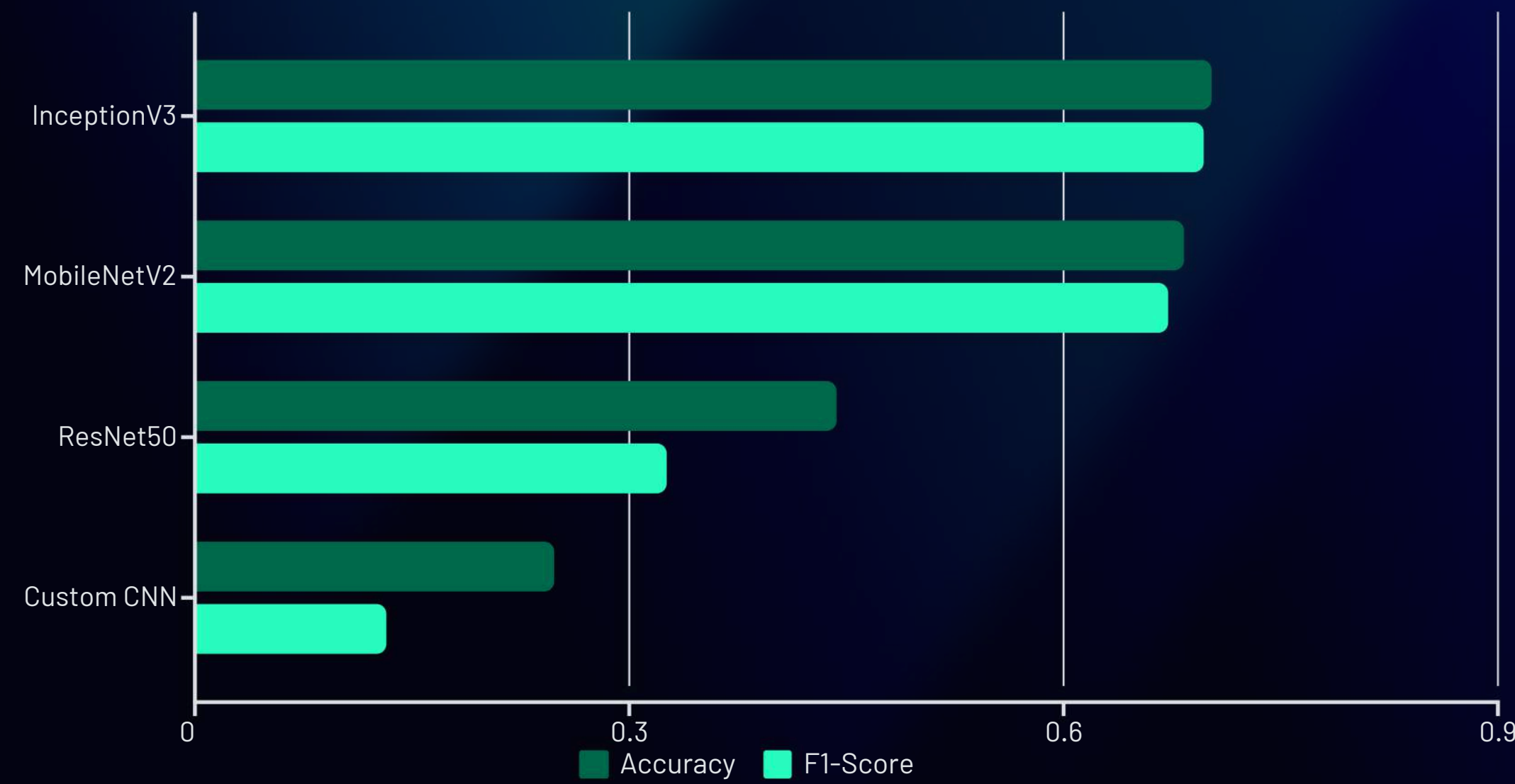
Models trained with **EarlyStopping** and **ModelCheckpoint** callbacks to monitor validation loss and prevent overfitting.

Architecture Highlights

Each model optimised for medical imaging tasks with balanced depth and computational efficiency for accurate tumour classification.

Model Evaluation

Comprehensive performance analysis across multiple deep learning architectures using standard classification metrics.



70.3%

BestAccuracy

InceptionV3 achieved highest performance

0.719

TopPrecision

InceptionV3 precision score

0.697

F1-Score

Balanced performance metric

Key Insight: InceptionV3 demonstrated the most balanced precision, recall, and F1-score, making it the optimal choice for deployment. Accuracy vs. loss curves plotted during training validated model convergence.

Streamlit Deployment



Upload

User uploads MRI scan image (.jpg/.png)



Preprocess

Image resized and normalised automatically



Predict

Model classifies tumour type with confidence



Show Results

Display predictions with confidence scores

App Features

- Upload MRI image for instant analysis
- Model predicts tumour type with confidence score
- Displays confidence bar chart for all four classes
- User-friendly, responsive interface with theme styling

Prediction Classes

- Glioma
- Meningioma
- Pituitary
- No Tumour



Platform: Streamlit App | **Title:** Brain Tumour Classifier |
 Simple, interactive, and lightweight UI with built-in "About" section

Conclusion & Future Scope

Key Achievement

Successfully classified brain MRI images into four tumour types using deep learning. **InceptionV3 achieved 70.3% accuracy** with the most balanced precision, recall, and F1-score.

Deployment Success

Streamlitwebappenables real-time MRI image uploads and displays instant tumour-type predictions with confidence scores for clinical support.

Current Limitations

Computational Constraints: Models trained using Google Colab CPU, restricting capacity and training speed.

Future Enhancements



GPU/TPU Training

Re-trainmodelsusing GPU or TPU to reduce training time and enhance accuracy significantly.



3D MRI Volume Analysis

Extend the system to process 3D volumetric MRI data instead of 2D slices, enabling more accurate tumor localization and segmentation.



Tumour Segmentation

IntegrateGrad-CAM andsegmentation modules for visual tumour localisation along with classification.



Dataset Expansion

Expand datasetsize and diversity for better model generalisation across varied patient populations.



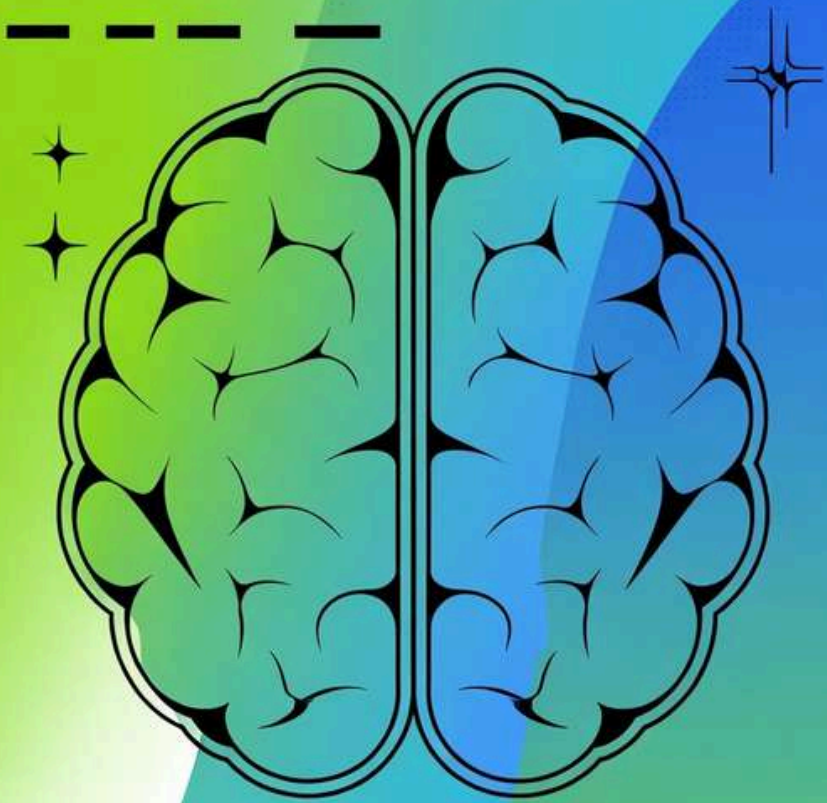
Cloud Deployment

DeployonAWS/GCP forreal-time hospital or telemedicine use with scalable infrastructure.



Edge Computing

Explorelightweightedge-compatible models for mobile and point-of-care diagnostic applications.



Thank You!

Brain Tumour MRI Image Classification Project

Empowering medical diagnosis through artificial intelligence